

# LTIMB10-D1: 25.11.2025 - DWH Concepts

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## 1. Introduction to Data Warehousing(DWH)

A **Data Warehouse (DWH)** is a *central storage system* where a company keeps all its historical data for **reporting, analytics, dashboards, and decision-making**.

Think of it like:

- ☐ **A big library** storing years of information from different departments (Sales, Marketing, Finance, HR, etc.)
  - ☐ Cleaned, organized, and structured for analysis.
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## 2. What is a Data Warehouse?

A **Data Warehouse** is:

A centralized system used for storing large volumes of historical data from different sources, optimized for fast analysis and reporting.

**Key points:**

- Stores **historical** data (1 year, 5 years, 10 years...)
  - Data is **cleaned, transformed, and organized**
  - Used for **BI Reports, Dashboards, Trends, Forecasting**
  - Not used for daily transactions
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### 3. OLTP vs. OLAP

| Feature         | OLTP                                      | OLAP                                 |
|-----------------|-------------------------------------------|--------------------------------------|
| Meaning         | Online Transaction Processing             | Online Analytical Processing         |
| Purpose         | Run daily business operations             | Analyze data for decisions           |
| Example         | Place order, transfer money, add customer | Monthly sales report, trend analysis |
| Data stored     | Current data                              | Large historical data                |
| Operations      | Insert, update, delete                    | Aggregation, grouping, reporting     |
| Speed target    | Fast transactions                         | Fast queries                         |
| Example Systems | Banking system, E-commerce website        | Data Warehouse, Power BI, Tableau    |

#### Simple Example (Uber/Ola)

- **OLTP:** When a user books a cab → data stored immediately
- **OLAP:** Company checks
  - which city got most bookings,
  - peak time analysis,
  - driver performance report

**OLTP runs the business.**

**OLAP helps *analyze* the business.**

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## 4. Benefits and Use Cases of DWH

### ✓ Benefits of DWH

1. **Single source of truth**
    - All company data stored in one place
  2. **Better decision-making**
    - Managers see accurate reports
  3. **Fast reporting**
    - Optimized for BI tools (Power BI, Tableau)
  4. **Stores years of history**
    - Useful for forecasting
  5. **Improves data quality**
    - Duplicate removal, standardization, consistency
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### ✓ Use Cases of Data Warehouses (Industry Examples)

#### 1. Retail (Amazon, Walmart)

- Best-selling products
- Inventory forecasting
- Customer buying trends

#### 2. Banking

- Fraud analysis
- Risk modeling
- Customer segmentation

### **3. Healthcare**

- Patient history analytics
- Treatment effectiveness trends

### **4. Telecom**

- Network usage reports
- Customer churn prediction

### **5. E-commerce**

- Daily sales dashboard
  - Cart abandonment analysis
  - Recommendation systems
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# 5. Key Components of a Data Warehouse System

Here are the building blocks of a DWH:

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## 1. Data Sources

Where data comes from:

- OLTP databases
  - CRM (Salesforce)
  - ERP (SAP)
  - Excel/CSV files
  - API data
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## 2. ETL/ELT Layer

ETL = Extract → Transform → Load

ELT = Extract → Load → Transform

Tools:

- Informatica
- SSIS
- Matillion
- Talend
- dbt

Purpose:

- collect data
  - clean it
  - standardize
  - remove duplicates
  - load into DWH
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### 3. Staging Area

Temporary storage for raw data

→ Dirty data lives here before cleaning

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### 4. Data Warehouse Storage

Central data store

Models used:

- **Star schema**
- **Snowflake schema**
- **3NF (normalized)**

Tables:

- **Fact tables** (numbers: sales, revenue)
  - **Dimension tables** (details: customer, products)
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### 5. Data Marts

Small, department-wise mini-warehouses

- Sales mart
  - Finance mart
  - HR mart
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### 6. BI/Analytics Layer

Tools for reporting:

- Power BI
  - Tableau
  - Looker
  - Qlik
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## 7. Metadata & Governance

Information about data:

- who owns it
- quality
- data definitions

Governance ensures:

- security
  - access control
  - compliance
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## 6. Introduction to Snowflake (Cloud Data Warehouse)

**Snowflake** is a **fully cloud-based data warehouse** — secure, scalable, and extremely fast.

### ✓ Key Features (Simple terms)

#### 1. Separation of Storage & Compute

- Store unlimited data
- Add unlimited computing power separately

#### 2. Auto-scaling

- Handles small & large workloads automatically

#### 3. Zero-copy cloning

- Create an instant copy of your database without extra storage

#### 4. Time Travel

- See data as of yesterday, last week, or last month

## 5. **Built-in security**

→ Role-based access, encryption, masking

## 6. **Works with Structured + Semi-Structured Data**

→ JSON, XML, Parquet all supported

## ✓ **Real Example (E-commerce company using Snowflake)**

- Sales data coming every minute
- Product catalog updated daily
- Customer clicks stored as JSON
- Finance team loads monthly invoices

Snowflake stores everything and serves:

- Sales dashboards
  - Forecasting reports
  - Customer behavior analysis
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# 7. Data Warehouse Lifecycle

This is how a DWH project happens in real companies:

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## 1. Requirement Gathering

- What reports needed?
- What KPIs matter?
- Who are stakeholders?

### Example:

Marketing team wants monthly customer purchase trend.

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## 2. Data Source Analysis

Understand where data resides

- SQL database
  - SAP
  - Flat files
  - APIs
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## 3. Data Modeling

- Fact tables
- Dimension tables
- Star schema

### Example:

Create:

Fact\_Sales

Dim\_Customer

Dim\_Product

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## 4. ETL Development

Build pipelines to:

- Extract data
- Clean it
- Transform
- Load into warehouse

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## 5. Testing

Types:

- Data validation testing
- Performance testing
- Load testing

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## 6. Deployment

Move to production environment

- schedule jobs
- set up monitoring

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## 7. Maintenance & Enhancements

- Add new columns
  - Update logic
  - Add more data sources
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# Summary Table for Quick Understanding:

| Topic           | Simple Meaning                                       |
|-----------------|------------------------------------------------------|
| DataWarehouse   | Central storage for analysis                         |
| OLTP            | Daily transactions                                   |
| OLAP            | Reporting & analytics                                |
| ETL/ELT         | Moving & transforming data                           |
| Fact table      | Contains numeric values                              |
| Dimension table | Contains descriptive data                            |
| Snowflake       | Cloud-based DWH                                      |
| DWH Lifecycle   | Steps: Requirement → Design → ETL → Testing → Deploy |

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